

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics	COURSE CODE: CSA1501
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I P.Poorna Chandra (192472287)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Poorna Chandra

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications. Analyze performance, security, and manageability, then justify the better choice.

Answer

Feature	Type-1 Hypervisor (Bare Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on hardware	Runs on top of an operating system
Performance	Very high with low overhead	Lower due to host OS overhead
Security	More secure because there is no host OS	Less secure since attacks on host OS can affect VMs
Manageability	Centralized management and live migration	Suitable for small-scale testing only
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Analysis

- **Performance:** Type-1 hypervisors directly access hardware, giving faster execution and lower latency.
- **Security:** Since no host operating system exists, the attack surface is smaller.
- **Manageability:** Supports centralized monitoring, VM migration, high availability, and disaster recovery.

Justification

For **mission-critical cloud applications**, **Type-1 Hypervisor** is the better choice because it offers:

- High performance
- Better security
- High availability
- Easy centralized management
- Better scalability

2. A Cloud Provider Must Isolate Workloads Belonging to Finance, Healthcare, and Student Portals on the Same Hardware. Analyze How Virtualization Architecture Supports Isolation and Identify Two Major Attack Surfaces.

Answer

Virtualization Architecture for Isolation

Virtualization allows multiple Virtual Machines (VMs) to run independently on the same physical server.

Each department (Finance, Healthcare, Student Portal) gets:

- Separate virtual CPU
- Separate virtual memory
- Separate storage
- Separate virtual network

Even if one VM fails or is compromised, the others remain unaffected.

Benefits

- Strong workload isolation
- Better resource utilization
- Improved security
- Easy management

Major Attack Surfaces

1. Hypervisor Attack

- If an attacker compromises the hypervisor, all VMs can be affected.
- Protection:
 - Secure hypervisor updates
 - Access control
 - Multi-factor authentication

2. VM Escape Attack

- A malicious VM escapes its isolation and accesses another VM or the host.
- Protection:
 - Regular patching
 - Secure VM configuration
 - Network segmentation

Conclusion

Virtualization provides secure isolation while efficient security practices protect against hypervisor attacks and VM escape.

3. Study the Following VM Host Utilization Snapshot and Recommend Corrective Actions: CPU 92%, Memory 88%, Storage I/O Latency 35 ms, Average VM Boot Time 170 s. Explain the Likely Bottlenecks.

Answer

Analysis of Resource Usage

Resource	Usage	Analysis
CPU	92%	CPU is overloaded
Memory	88%	Memory is almost full
Storage I/O Latency	35 ms	High disk delay
VM Boot Time	170 s	Slow VM startup

Likely Bottlenecks

CPU Bottleneck

- Too many VMs share CPU resources.
- Results in slow processing.

Memory Bottleneck

- Insufficient RAM causes swapping.
- Performance decreases.

Storage Bottleneck

- High I/O latency delays disk operations.
- Applications and VM startup become slow.

Corrective Actions

1. Add more CPU cores.
2. Increase RAM capacity.
3. Upgrade HDD to SSD/NVMe storage.
4. Balance VM workloads across hosts.
5. Remove unused VMs.
6. Enable resource monitoring and auto-scaling.

Conclusion

The major bottlenecks are **high CPU utilization, insufficient memory, and slow storage performance**. Upgrading hardware and balancing workloads will improve VM performance.

4. Analyze How Software Defined Datacenter (SDDC) Architecture Improves Agility Compared with a Traditional Datacenter. Support Your Answer with Compute, Storage, and Network Virtualization Considerations.

Answer

Traditional Datacenter

- Hardware is configured manually.
- Resource allocation is slow.
- Scaling requires physical installation.
- High operational cost.

Software Defined Datacenter (SDDC)

All infrastructure resources are controlled through software.

Compute Virtualization

- Creates virtual machines quickly.
- Supports live migration.
- Enables automatic scaling.

Storage Virtualization

- Combines multiple storage devices into one logical pool.
- Simplifies storage management.
- Improves backup and disaster recovery.

Network Virtualization

- Creates virtual networks.
- Enables software-controlled routing and firewall policies.
- Faster network deployment.

Advantages of SDDC

- Faster deployment
- Better scalability
- Automation
- Reduced operational cost
- Improved flexibility
- Efficient resource utilization

Conclusion

SDDC provides greater agility than traditional datacenters by virtualizing compute, storage, and networking, enabling rapid deployment and centralized management.

5. A Multi-Cloud Federation Project Fails Due to Identity Mismatches Between Providers. Analyze the Issue and Propose a Secure Identity and Privacy Design Using Federation Concepts.

Answer

Issue Analysis

Different cloud providers use different identity systems.

Problems include:

- Different user IDs
- Different authentication methods
- Duplicate user accounts
- Access control inconsistencies

These issues prevent users from accessing resources across multiple clouds.

Secure Identity Federation Design

Identity Provider (IdP)

- Maintains one central user identity.

Service Providers (SPs)

- Trust the Identity Provider.
- Users log in once and access multiple cloud services.

Single Sign-On (SSO)

- One login provides access to multiple cloud platforms.

Federation Standards

- SAML
- OAuth 2.0
- OpenID Connect

Privacy Protection

- Encrypt user information.
- Share only required user attributes.

- Apply role-based access control (RBAC).
- Enable Multi-Factor Authentication (MFA).
- Maintain audit logs.

Conclusion

Using identity federation with **SAML/OpenID Connect, Single Sign-On, MFA, and Role-Based Access Control** provides secure authentication, protects user privacy, and solves identity mismatches across multiple cloud providers.